

Project 1: Explainable Machine Learning for Gallstone Disease

Data type: Tabular

Gallstone disease is a common medical condition that can be associated with demographic, metabolic, body-composition, and biochemical characteristics. The dataset contains information about patients with and without gallstones, including demographic variables, body measurements, bioimpedance measurements, hepatic fat measurements, and laboratory test results. Machine-learning models can potentially identify patterns associated with gallstone disease. However, simply obtaining a high classification accuracy does not tell us why a model makes a particular prediction.

Dataset: Link to dataset: <https://archive.ics.uci.edu/dataset/1150/gallstone-1>

- Cohort & timeframe: 319 outpatients from Ankara VM Medical Park (June 2022–June 2023); 161 diagnosed with gallstones; ethics approval: Ankara City Hospital (E2-23-4632).
- Features (38 total): demographics (age, sex, height, weight, BMI); bioimpedance (total/extracellular/intracellular water, muscle/fat mass, protein, visceral fat area, hepatic fat); labs (glucose, lipids, AST/ALT/ALP, creatinine, GFR, CRP, hemoglobin, vitamin D). The dataset has no missing values and has balanced classes.

The objective of this project is therefore twofold:

1. **Develop a machine-learning model that can distinguish patients with and without gallstone disease.**
2. **Investigate which clinical characteristics contribute most to the model's predictions.**

Reference

Introductory Paper: <https://www.ovid.com/jnls/md-journal/fulltext/10.1097/md.00000000000037258~early-prediction-of-gallstone-disease-with-a-machine>

Project 2: Anomaly Detection in Oil Well Production Time-Series Data

Data type: Tabular + Time series

Anomaly detection in industrial time-series data is important for identifying abnormal operating conditions and potential failures in oil and gas production systems. The **Petrobras 3W dataset** provides multivariate time-series data from oil wells, including normal operating conditions and several types of production-related anomalies.

Previous research has explored a wide range of approaches, including traditional machine-learning methods such as Random Forest, Decision Trees, and XGBoost, as well as deep learning approaches based on LSTM, GRU, convolutional neural networks, and autoencoders. Both supervised and unsupervised/semi-supervised approaches can be relevant, depending on how the available labels are used.

Develop and evaluate a machine-learning approach for detecting anomalous behaviour in oil-well production time-series data using the Petrobras 3W dataset.

Dataset: Link to dataset <https://github.com/petrobras/3W>

References

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- Matheus A. Marins, Bettina D. Barros, Ismael H. Santos, Daniel C. Barrionuevo, Ricardo E.V. Vargas, Thiago de M. Prego, Amaro A. de Lima, Marcello L.R. de Campos, Eduardo A.B. da Silva, and Sergio L. Netto. Fault detection and classification in oil wells and production/service lines using random forest. *Journal of Petroleum Science and Engineering*, 197:107879, 2021. doi: 10.1016/j.petrol.2020.107879.
 - Roel Bouman and Tom Heskes. Autoencoders for anomaly detection are unreliable. *arXiv preprint arXiv:2501.13864*, 2025
 - Qihang Zhou, Guansong Pang, Yu Tian, Shibo He, and Jiming Chen. AnomalyCLIP: Object-agnostic prompt learning for zero-shot anomaly detection. In *The Twelfth International Conference on Learning Representations (ICLR)*, 2024
 - Mohit Kakda, Mirudula Shri Muthukumaran, Uttapreksha Patel, and Lawrence Swaminathan Xavier Prince. Analyzing VLM-based approaches for anomaly classification and segmentation. *arXiv preprint arXiv:2601.13440*, 2026
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Project 3: Machine Learning for Fatigue Lifetime Prediction of Structural Steels

Data type: Tabular + regression

Fatigue failure is a major cause of structural failure in mechanical components. The goal of this project is to develop machine-learning models that predict the fatigue lifetime of structural steels from material properties and experimental loading conditions. Students will work with experimental fatigue data that include steel/material properties, mechanical characteristics, loading conditions, and fatigue-test parameters. The target variable will be the number of cycles to failure.

Main research question: Can machine learning accurately predict the fatigue lifetime of structural steels from material properties and experimental loading conditions?

Dataset: Link to dataset : Go to Canvas, Files, Datasets, [Project 3](#)

References

- Konstantinos Arvanitis; Pantelis Nikolakopoulos; Dimitrios Pavlou; Mina Farmanbar (2025) Machine learning-based fatigue lifetime prediction of structural steels. I: Alexandria Engineering Journal. ISSN 1110-0168. Volum 125. s.55-66. DOI: 10.1016/j.aej.2025.04.014
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ADVANCED TOPICS

Project 4: Detecting Dementia Using Natural Language Processing

Data type: Text + NLP

This project aims to develop a predictive model for detecting dementia using Natural Language Processing (NLP). The model will utilize linguistic features extracted from text data to predict dementia diagnoses. You will be responsible for preprocessing the data, performing exploratory analysis, and building machine learning or deep learning models to classify transcripts into either dementia or control categories (binary classification). Students are encouraged to explore model optimization and fine-tuning techniques using frameworks like Scikit-learn, TensorFlow, Keras, Transformer or PyTorch.

Dataset: The dataset for this project is sourced from DementiaBank. The transcript files have been prepared in CSV format and divided into training and testing sets:

- Training Dataset: Includes two files, Dementia_db.csv (containing dementia patient transcripts) and Control_db.csv (containing healthy control transcripts).
- Testing Dataset: Contains one file, testing_db.csv, for evaluating the model's performance.

Link to dataset: Go to Canvas, Files, Datasets, [Project 4](#)

References:

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- Mina Farmanbar; Shaima Ahmad Freja; Arezo Shakeri (2025) Improving Early Dementia Detection with Advanced Language Models Based on Linguistic Features. I: Klaus Schoeffmann; Ajit Kumar Verma; Nikolaos Kouvakas; Maria Tzamtzi; Mina Farmanbar, Horizons of AI : Ethical Considerations and Interdisciplinary Engagements. *Springer Nature*. ISBN 9789819679447. s.161-175. DOI: 10.1007/978-981-96-7945-4_11
 - Arezo Shakeri; Mina Farmanbar; Krisztian Balog (2025) MultiConAD: A Unified Multilingual Conversational Dataset for Early Alzheimer's Detection. I: Omar Alonso; Gabriella Pasi; Nicola Ferro; Andrew Trotman; Suzan Verberne; Maria Maistro, SIGIR '25: The 48th International ACM SIGIR Conference on Research and Development in Information Retrieval. *Association for Computing Machinery (ACM)*. ISBN 9798400715921. s.3627-3637. DOI: 10.1145/3726302.3730313
 - Arezo Shakeri; Shaima Ahmad Freja; Yeganeh Hallaj; Mina Farmanbar (2024) Uncovering linguistic patterns : A machine learning exploration for early dementia detection in speech transcripts. I: Hasan Ogul; Jonas Aamodt Moræus, 2024 4th International Conference on Applied Artificial Intelligence (ICAPAI). *IEEE (Institute of Electrical and Electronics Engineers)*. ISBN 9798350349764. DOI: 10.1109/ICAPAI61893.2024.10541296

Data type: Image + Text

- number of words
- vocabulary
- repeated words
- sentence structure
- number of nouns and verbs
- pauses
- speech rate
- semantic similarity
- coherence



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The transcript tells us what the person said, but it does not directly tell us what the person failed to mention compared with what was present in the picture. A Vision-Language Model (VLM) can potentially examine both pieces of information: the original picture + the patient's description and compare them.

The main research question is: Can Vision-Language Models extract meaningful visual-semantic features from picture description tasks, and can these features help distinguish individuals with AD, MCI and normal cognition?

Dataset: Same as Project 4 dataset. Go to Canvas, Files, Datasets, [Project 4](#)

References

- Li, F., Luo, J., & Xia, W. (2025). WavFusion: Towards wav2vec 2.0 multimodal speech emotion recognition. In *Proceedings of the 31st International Conference on Multimedia Modeling (MMM 2025)*. Springer. https://doi.org/10.1007/978-981-96-2071-5_24
 - Duret, J., Rouvier, M., & Estève, Y. (2024). MSP-Podcast SER Challenge 2024: L'antenne du Ventoux multimodal self-supervised learning for speech emotion recognition. In *Proceedings of the Speaker and Language Recognition Workshop (Odyssey 2024)*. ISCA. arXiv:2407.05746
 - <https://www.tandfonline.com/doi/full/10.1080/13803395.2018.1446513>
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Project 6: What does a pretrained ECG model actually see?

You will pick a subset of PTB-XL (say 1,000–2,000 records - small enough to be comfortable on a free-tier T4 Colab GPU, large enough that the clustering result means something) and run it through one frozen pretrained ECG encoder to get embeddings. Then PCA plus at least one of t-SNE/UMAP, backed by an actual number (k-NN label agreement or silhouette score), not just a scatter plot and call it evidence.

The real ask is going past "yes it separates" into characterizing where it doesn't: which diagnostic pairs blur together, and whether that's consistent across methods.

Dataset: PTB-XL for signals and labels. <https://physionet.org/content/ptb-xl/1.0.3/>
https://physionet.org/content/ptb-xl/1.0.1/ptbxi_database.csv

Hints:

1. Use ECGFounder for the extraction, not ECG-FM, ECG-FM's own pretraining corpus includes PTB-XL, which would make any separation finding circular.
 2. ECGFounder's own public eval script only exposes classification logits, not embeddings. Getting a feature vector out means calling the underlying network with `return_features=True` and pulling the intermediate features instead of the class-probability output, the public script alone won't show them this.
 3. No need to download the full 1.7 GB PTB-XL signal zip for a 1–2k subset. wfdb's reader can stream individual records straight from PhysioNet's server with the `pn_dir` argument instead of pulling the whole database locally first.
 4. The encoder was trained with diagnostic labels, so class separation by itself isn't surprising. What's worth reporting is which classes blur together despite that supervision, and whether it lines up with clinical similarity (e.g. classes that look alike on paper) or looks arbitrary.
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References:

- Wagner, P., Strodthoff, N., Bousseljot, R., Samek, W., & Schaeffter, T. (2022). PTB-XL, a large publicly available electrocardiography dataset (version 1.0.3). PhysioNet. RRID:SCR_007345. <https://doi.org/10.13026/kfzx-aw45>
- Wagner, P., Strodthoff, N., Bousseljot, R.-D., Kreiseler, D., Lunze, F.I., Samek, W., Schaeffter, T. (2020), PTB-XL: A Large Publicly Available ECG Dataset. Scientific Data. <https://doi.org/10.1038/s41597-020-0495-6>
- Pollard, T., Moody, B. E., Lehman, L., Gow, B., Fernandes, C., Xie, C., Johnson, A., Mark, R. G., & Heldt, T. (2026). PhysioNet as a global platform for biomedical research. Nature Health. <https://doi.org/10.1038/s44360-026-00096-z>. Available from: <https://rdcu.be/faatM>

Project 7: Coronary artery segmentation from CCTA images

Data type: Image

Automated segmentation of blood vessels plays a vital role in diagnosis, surgical planning, and hemodynamic modeling of cardiovascular diseases. The coronary arteries that supply blood to the heart muscle are of particular interest since coronary artery disease (CAD) is the most common cardiovascular disease. CAD is characterized by the narrowing or blockage of coronary arteries and can lead to chest pain or a heart attack. The coronary arteries consist of two main branches, one on each side of the heart. They have a complex, tortuous structure with many small branches, and can look very different between patients. In this project, you will develop, train, and evaluate a deep learning pipeline to segment coronary arterial trees using the ImageCAS dataset of CCTA images. You will use the expert-annotated ImageCAS-X dataset (finished gold-standard segmentations of ImageCAS) to validate your segmentation model.

This project can be completed with two different tracks (choose one):

1. Foundational implementation

Objective: Implement a simple 2D or 3D segmentation network based on the U-Net architecture using PyTorch

Focus: Deep understanding of the fundamentals behind segmentation algorithms

2. Extend or tweak an existing segmentation model

Objective: Adapt an established medical image segmentation model to the coronary artery dataset and introduce a distinct algorithmic modification by extending or changing the existing model

Focus: Architectural modification of segmentation algorithms

Dataset:

ImageCAS paper: <https://www.sciencedirect.com/science/article/pii/S0895611123001052>

ImageCAS-X paper: <https://arxiv.org/abs/2608.30404>

CCTA image dataset (ImageCAS) available at

<https://www.kaggle.com/datasets/xiaoweixumedicalai/imagecas/data>

Segmented dataset (ImageCAS-X) and pretrained weights for some segmentation methods available at <https://zenodo.org/records/21887809>

References

Available segmentation methods to use as a baseline:

1. Swin-UNETR

- Paper: <https://arxiv.org/pdf/2201.01266v1>

- Code: <https://github.com/LeonidAlekseev/Swin-UNETR>

2. nnU-Net

- Paper: <https://www.nature.com/articles/s41592-020-01008-z>

- Code: <https://github.com/mic-dkfz/nnunet>

3. CAS-Net

- Paper: <https://www.sciencedirect.com/science/article/pii/S1361841523000063>
 - Code: <https://github.com/Cassie-CV/CAS-Net/tree/main>
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